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ONLINE CERTIFICATION COURSE ON

Statistical and Machine Learning Methods: Basics and Applications

MODULE 8 : Introduction to Statistical Learning

LECTURE W8L1: Introduction to Statistical Learning

BY

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CONCEPTS COVERED

Introduction to Statistical Learning

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Statistical Learning

- Statistical learning is a mathematical framework for understanding data. It blends traditional statistics with machine learning to model and understand complex datasets.
- It can be applied any field of studies. Often several variables are dependent on the set of independent variables and the association may be linear or non linear.
- Statistical learning is all about the set of methods used to investigate the association between the independent (input) and dependent (output) variables.
- If the association between the input and output variables were determined, the beneficial insights can be drawn from the dataset.

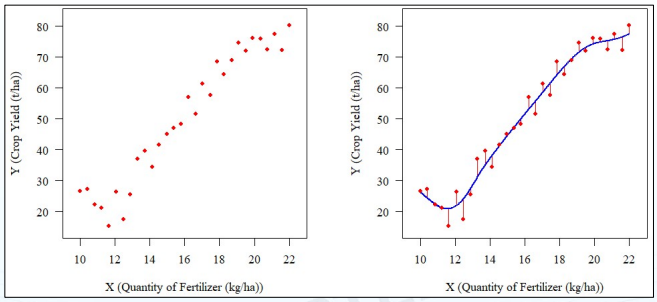
Statistical Learning

- Let us assume that we observe a quantitative response Y for p different predictors, $X_1, X_2, \dots, X_p$.
- There will be some relationship between Y and $X = (X_1, X_2, \dots, X_p)$, which can be written in general form,

$$Y = f(X) + \epsilon$$

- Here, f is some fixed but unknown function of $X_1, X_2, \dots, X_p$, and ε is a random error term, which is independent of X and has mean zero.
- In this formulation, f represents the systematic information that X provides about Y.

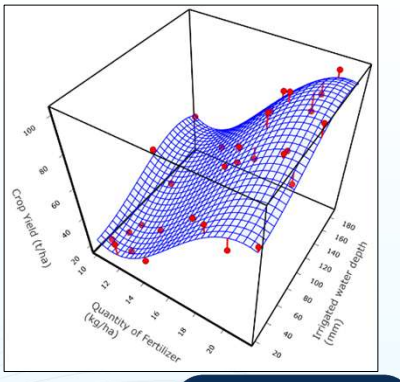
Statistical Learning



- The above plot shows the quantity of fertilizer applied (x-axis) and the corresponding crop yield (y-axis).
- The function f that connects the input variable to the output variable is, in general, unknown.
- Upon estimating the function f, the vertical lines represent the error terms $\in$.

Statistical Learning

- In general, the function f may involve more than one input variable.
- In the same way, we can visualize the same dataset with two dependent variables, for example, quantity of fertilizer, Irrigated water depth, and the response variable crop yield.
- Here, f is shown as the two-dimensional surface that must be estimated based on the observed data.
- Therefore, statistical learning refers to a set of approaches for estimating f.



Statistical Learning

There are two major verticals: Prediction and Inference

Prediction

If the set of inputs (X) is available, the output (Y) be estimated.

$$\hat{Y} = \hat{f}(X)$$

- Here $\hat{f}$ represents estimate for f, and $\hat{Y}$ represents the estimated value of Y, given the set of X.

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- The accuracy of $\hat{Y}$ as a prediction for Y depends on two quantities, i.e., reducible error and irreducible error.
- Since $\hat{f}$ is not a perfect estimate of f, the resulting inaccuracy introduces an error known as the reducible error.
- This error can be reduced by improving the accuracy of $\hat{f}$ through the selection of more appropriate techniques for estimating f.
- However, even if it were possible to form a perfect estimate for f, i.e., $\hat{Y} = f(X)$, the prediction would still possess error.
- This is due to the fact that Y is also a function of ϵ.
- This error is known as the irreducible error.

Statistical Learning

- Consider a given estimate $\hat{f}$ and set of predictors X which yields the prediction $\hat{Y} = f(X)$.
- Upon fixing the function $\hat{f}$ and X the variability comes from ϵ can be shown as

$$E\left(Y - \hat{Y}\right)^2 = E\left[f(X) + \epsilon - \hat{f}(X)\right]^2$$

$$E\left(Y - \hat{Y}\right)^2 = \underbrace{\left\{\left[f(X) - \hat{f}(X)\right]^2\right\}}_{Reducible} + \underbrace{\{Var(\epsilon)\}}_{Irreducible}$$

where $E\left(Y - \hat{Y}\right)^2$ represents the average, or expected value of the squared difference between the predicted and actual value of Y, and $Var(\epsilon)$ represents the variance associated with the error term ϵ.

Statistical Learning

- In case of inference, the interest lies in understanding the association between Y and $X_1, X_2, \ldots, X_p$, through the estimation of f in exact form, and the goal is not necessarily to make perfect predictions for Y.
- Interest lies in getting answer for the following questions:
  i. What are the predictors and corresponding response? Is there any relationship?
  ii. What are the underlying process(es)?
  iii. Is the data sufficient for the inference?

Statistical Learning

Estimation of f

- To estimate the unknown function f, the statistical learning methods are applied to the training data.
- The estimated function $\hat{f}$ such that $Y \approx \hat{f}(X)$ for any observation (X, Y).
- Broadly, these statistical learning methods for this task can be characterized as either parametric or non-parametric.

Parametric Methods: Parametric methods involve a two-step approach:

i. Assumption of functional form (linear, exponential, etc.)

ii. Estimation of parameters through training data.

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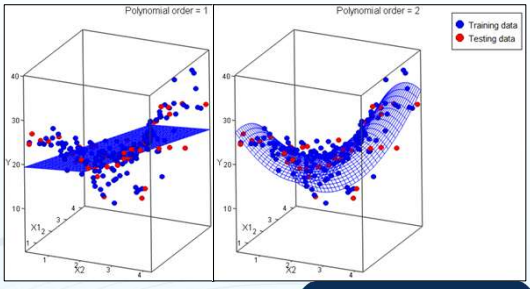
Non-Parametric Methods:

- Non-parametric models do not explicitly assume the functional form of f.
- They have the potential to accurately fit a wider range of possible shapes for f.
- These models make no assumptions about the intricate patterns of the input and output variables.
- However, non-parametric models require a very large number of observations, in order to obtain an accurate estimation for f.

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Trade-off: Prediction Accuracy and Model Interpretability

- Linear regression is a relatively inflexible approach, because it can only generate linear functions.
- Non-parametric methods are considerably more flexible.
- The choice between the flexible and inflexible approaches is dependent on the objective, i.e., inference (interpretability) or prediction.



Trade-off: Prediction Accuracy and Model Interpretability

- If the inference is the goal, then a linear model may be a good choice because it is relatively easy to understand the relationship between the predictor and the predictand set.
- In case of highly flexible approaches, it can produce complicated estimates of f, at the same time, making it difficult to understand how any individual predictor is associated with the response.
- Therefore, when inference is the primary goal, there are clear advantages in using relatively inflexible statistical learning methods.
- If prediction is the primary goal, more flexible models may improve prediction accuracy but are also more prone to overfitting.

Example-1 (illustrative)

Let x be an independent variable and let the dependent variable y vary according to the quadratic equation:

$$y = x - 0.2x^2$$

The response ($\hat{y}$) data consists of true signal (y) and random noise (ϵ),

$$\hat{y} = y + \epsilon,$$

Let the polynomial be of order m:

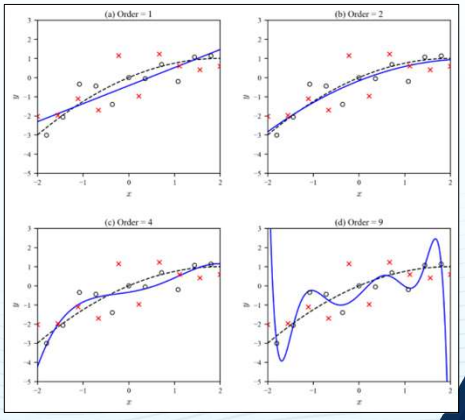
$$\hat{y} = \theta_0 + \theta_1 x + \theta_2 x^2 + \cdots + \theta_m x^m$$

where θ_j $(j = 0, \ldots, m)$ are the $m + 1$ adjustable parameters or weights, and $\hat{y}$ denotes the predicted output of the polynomial function.

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Example-1 (illustrative)

- Polynomials of order 1, 2, 4 and 9 are fitted to the training dataset by minimizing the *mean squared error (MSE)* between the model output ($\hat{y}$) and the true data (y).
- On comparison, it is observed that there is an improvement in the agreement of polynomial fit with y from order 1 to order 2.
- The agreement gets worse at higher-order polynomials.

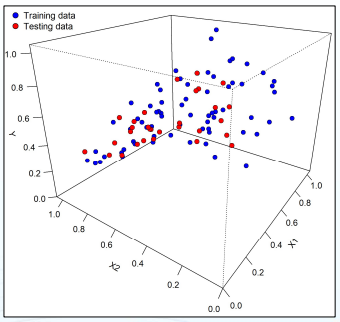


Example-1 (illustrative)

- The above illustration shows the concepts of underfitting and overfitting data.
- At polynomial order 1, the model is underfitting data since the y is quadratic, but the model is linear.
- However, polynomial orders 4 and 9 overfitted the data, with more adjustable parameters, which are greater than the parameters needed to fit the y.
- On testing with new test data (red cross), the polynomial order 9 curve predicts poorly, although an excellent fit is shown with the training data (black circles)
- In real-world applications, the true function y data is generally unknown i.e., it is not possible to directly determine whether a model is underfitting or overfitting.
- Therefore, a subset of testing data known as the validation dataset is used to test the model.

Example-2 (illustrative)

The dataset contains two predictor variables, X_1 and X_2, and one response variable, Y, as shown in the three-dimensional scatter plot. The data are divided into training and testing sets.



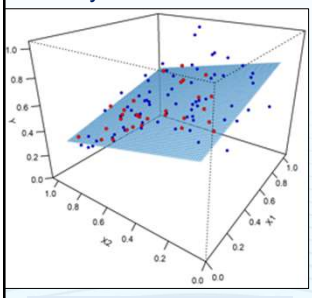
- Fit linear and nonlinear regression (polynomial) models to the given three-dimensional dataset.
- Visualize the fitted surfaces and compare the training and testing RMSE values to examine the effects of model flexibility, underfitting, and overfitting.
- The underlying true relationship as follows,

$$y = 3 + 1.5X_1 - 1.2X_2 + 2.5X_1^2 - 1.8X_2^2 + 1.5(X_1 \times X_2) + 0.8X_1^3 - 0.6X_2^3$$

Solution (illustration)

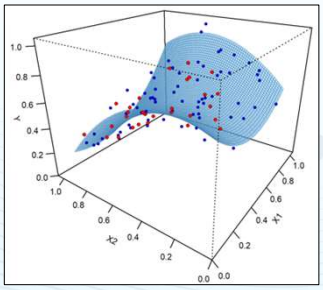
- The given data with two predictors and a response variable is fitted with different polynomial orders.
- The fitted results are as follows:

Polynomial order: 1



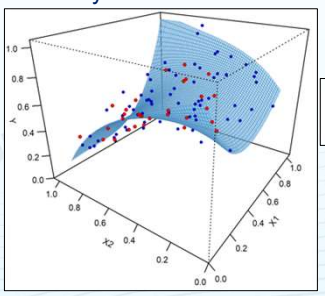
Training RMSE:5.63
Testing RMSE:7.34
Underfitting

Polynomial order: 2



Training RMSE:1.63
Testing RMSE:0.58
Best Model

Polynomial order: 3



Training RMSE:1.40
Testing RMSE:4.86
Overfitting

Solution (illustrative)

- The first-order model is too simple to capture the non-linear relationship and therefore shows underfitting.
- The second-order model gives the lowest training and testing RMSE, indicating the balance between model flexibility and predictive accuracy.
- Increasing the polynomial order beyond the second degree (greater than the number of parameters to predict the y_{true}) does not necessarily improve the performance.
- Although greater model flexibility can reduce training error, it may increase testing error due to overfitting.
- Therefore, for the given dataset, the second-order polynomial model is the best model based on its minimum testing $RMSE$ magnitudes.


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Statistical and Machine Learning Methods: Basics and Applications

MODULE 8 : Statistical Learning

LECTURE W8L2: Supervised and Unsupervised Learning

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CONCEPTS COVERED

Supervised and Unsupervised Learning

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Supervised and Unsupervised Learning

- Learning from the data: There are two main types of learning, supervised or unsupervised learning.
- Supervised learning is a type of machine learning where a model is trained on a labeled dataset.
- An analogy for supervised learning is a student trying to learn with an answer key for every question provided to him/her, i.e., the desired output (or label) is already provided by the teacher.
- In unsupervised learning an algorithm is trained on an unlabeled dataset. There is no explicit instructions. It is left on its own to explore the data and discover hidden patterns, groupings, or relationships etc.
- There is another learning, known as reinforcement learning, that is different from both supervised and unsupervised learning.

Supervised Learning

- Supervised learning tries to find a mapping from the input variables ($\boldsymbol{X}$) to the output variables (Y), i.e., observed or target data.
- Supervised learning is fundamentally divided into two categories: regression and classification.
- The broad distinction is made based on the type of output variables.
- In case of regression type models, target variable is continuous real valued.
- In case of classification, the target variables are either discrete or categorical.
- In regression, input variables are real-valued, although it is also possible to incorporate discrete or categorical variables as part of the inputs.

Supervised Learning

- Classification deals with discrete or categorical outputs $\hat{y} \epsilon \{1, \ldots, C\}$, where C represents the total number of target classes.
- When there are only two categories to classify, it is called a binary classification., otherwise more than two classes, generally known as multiclass classification.
- The classification task is more applicable in a vast number of applications, such as handwriting detection, object recognition and etc.,
- In some applications, the problems may utilize both the classification techniques sequentially.
- For example, to predict the streamflow in the next month, a classification model is often used first to determine whether it will be a high or low flow month, followed by a regression model to predict the approximate flow.

Unsupervised Learning

- In contrast to supervised learning, unsupervised learning works exclusively with input data ($\boldsymbol{X}$) with the goal of finding inherent structures within it.
- The goal here is to search for a common attribute among the given data, such as distance, spatio-temporal range of values, magnitudes of input data variables, etc.
- Clustering analysis: When data points are similar and located close together, clustering is used to group them.
- Another application involves reducing the high-dimensional input variable space by identifying the directions that display the strongest data spread.
- Principal Component Analysis (PCA) is commonly used to condense high-dimensional datasets, which can be considered as an unsupervised learning model.

Reinforcement Learning

- Reinforcement Learning (RL) represents a specialized branch of machine learning, frequently applied in many fields of studies.
- In RL, the model learns through sequential decisions by interacting with an environment.
- Unlike supervised learning, which relies on a static dataset of labeled examples, RL focuses on goal-directed learning from interaction, learning what to do to maximize a numerical reward signal.
- The reinforced learning model may evolve to outperform and surpass the target prediction skill through self-play and continuous adaptation through rewards and penalties.
- Predominantly applicable in improving the behavior of intelligent agents (e.g., robots), etc.

Comparison of Supervised and Unsupervised learning

Supervised Learning	Unsupervised Learning
Training required using labeled data	Training required using unlabeled data
Performance is checked using direct feedback from the available data	No feedback is used to check model performance
The training is done to develop the model for new data	The training is done to find the hidden patterns and useful insights
Two Broad categories of problems are Classification and Regression	Two broad categories of problems are Clustering and Association

Comparison of Supervised and Unsupervised learning

Supervised Learning	Unsupervised Learning
More accurate results are expected as we train the model for each data/pattern.	It may provide less accurate results as compared to supervised learning.
Example includes – • Linear Regression • Logistic Regression • Bayesian Models • Multi-class Classification • Artificial Neural Network • Support Vector Machine • K-Nearest Neighbour (KNN) • Decision tree, etc.	Example includes – • Clustering • Association model





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Statistical and Machine Learning Methods: Basics and Applications

MODULE 8 : Statistical Learning

LECTURE W8L3: Model Performance Evaluation

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CONCEPTS COVERED

Model Performance Evaluation

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Model Performance Evaluation

- Performance evaluation of a model is the process of quantifying the quality of model outputs against actual observations.
- A wide range of statistical metrics exists for this purpose, which are used for the assessment of prediction performance and forecast verification.
- However, the selection of a metric depends on the type of predictions/forecasts, i.e., whether it is numerical values, categorical values etc.
- Sometimes a combination of visual inspection and metric computation is necessarily adopted.
- The performance evaluation metrics can be divided into two groups: Numerical and Categorical.

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Performance Evaluation Metrics: Numerical Outputs

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Pearson's Correlation Coefficient

- Pearson's correlation coefficient, often called the correlation coefficient, which is also known as Pearson's product-moment correlation coefficient.
- It is a measure of linear association between two variables (as discussed in linear regression).
- Let r be the correlation coefficient,

$$r = \frac{\sum_{i=1}^{n}(O_i - \bar{O})(P_i - \bar{P})}{\sqrt{\sum_{i=1}^{n}(O_i - \bar{O})^2}\sqrt{\sum_{i=1}^{n}(P_i - \bar{P})^2}}$$

where, $O_i(i = 1,2,\dots,n)$ is the observed values and $P_i(i = 1,2,\dots,n)$ are the corresponding model outputs.

Pearson's Correlation Coefficient

- In two more ways, Pearson's correlation coefficient can be expressed by rearranging the terms.

$$r = \frac{\sum_{i=1}^{n} O_i P_i - n\bar{O}\bar{P}}{\sqrt{\sum_{i=1}^{n} O_i^2 - n\bar{O}^2}\sqrt{\sum_{i=1}^{n} P_i^2 - n\bar{P}^2}}$$

$$r = \frac{n\sum_{i=1}^{n} O_i P_i - \sum_{i=1}^{n} O_i^2 \sum_{i=1}^{n} P_i^2}{\sqrt{n\sum_{i=1}^{n} O_i^2 - \left(\sum_{i=1}^{n} O_i\right)^2}\sqrt{n\sum_{i=1}^{n} P_i^2 - \left(\sum_{i=1}^{n} P_i\right)^2}}$$

- The value of correlation coefficient ranges from -1 to $+1$, while -1 indicates a perfect negative linear association, $+1$ indicates a positive linear association.
- A value of 0 indicates the absence of any linear association

Adjusted Correlation Coefficient

- The concept of adjusted correlation is introduced to take care of the fact that the standard Correlation Coefficient is not an unbiased estimate of the corresponding population parameter (ρ), i.e.,

$$\rho = \frac{\sigma_{X,Y}}{\sigma_X \sigma_Y}$$

- The approximate unbiased estimator can be obtained through the adjusted correlation coefficient (r_{adj})

$$r_{adj} = \sqrt{1 - \left(\frac{(1 - r^2)(n - 1)}{n - 2}\right)}$$

The r_{adj} is approximately same as r for large value of n.

Spearman's Rank Correlation Coefficient

- The Spearman's rank correlation coefficient, also known as Spearman's correlation coefficient or Spearman's ρ, is the Pearson correlation coefficient between the rank values of two variables.
- In other words, Pearson's correlation coefficient is calculated using the raw values of the variables involved, but Spearman's correlation coefficient uses the corresponding ranks (1st, 2nd, 3rd, etc.) of the variables involved.
- The equation is similar to the Pearson correlation coefficient but expresses the notation by the ranks.

$$r_s = \frac{\sum_{i=1}^{n}\left(O_i^R - \bar{O}^R\right)\left(P_i^R - \bar{P}^R\right)}{\sqrt{\sum_{i=1}^{n}\left(O_i^R - \bar{O}^R\right)^2}\sqrt{\sum_{i=1}^{n}\left(P_i^R - \bar{P}^R\right)^2}}$$

The superscript R is used to indicate that these are rank values.

Spearman's Rank Correlation Coefficient

- If there are no ties and all ranks are distinct values, the simplified formula is expressed as,

$$r_s = 1 - \left(\frac{6\sum_{i=1}^{n} d_i^2}{n(n^2-1)}\right)$$

where, d_i is the difference between the ranks of ith pair of observed and corresponding model output.

- The Spearman's correlation coefficient indicates the existence of monotonic association that may or may not be linear.
- Spearman's correlation coefficient also ranges between +1 and -1.

Spearman's Rank Correlation Coefficient

- In case of a perfect monotone association between observed and model output, and there are no ties, Spearman correlation is equal to +1 or -1, depending on a positively or negatively monotone association.
- Another important point related to **Spearman's rank correlation coefficient is –**
  - higher values of Spearman's correlation coefficient not necessarily indicate a better model performance, rather there is some relationship still exist that can be incorporated in the model so that a better performance can be obtained, particularly in the tail ends (extreme values).

Kling Gupta Efficiency (KGE)

- The KGE is formulated by expressing model performance as a function of three components: linear correlation, variability bias, and bias in the mean.
- The correlation term represents how well the model captures the temporal variations of the observed data.
- The variability ratio compares the standard deviation of model output with that of observed data and shows how well the model captures the variability of the data.
- The bias ratio compares the mean of the model output with the mean of observed data and indicates whether the model underestimates or overestimates the observations.

Kling Gupta Efficiency (KGE)

- By considering the components (linear correlation, variability bias, and bias in the mean) together, KGE is used to evaluate the model performance.

$$KGE = 1 - \sqrt{(r-1)^2 + (\beta-1)^2 + (\gamma-1)^2}$$

where, r is the correlation output between the model output and observed values. γ and β are the variability ratio and bias ratio respectively.

- The bias ratio (β) is expressed as, $\beta = \frac{\mu_p}{\mu_0}$
- The variability ratio (γ) is expressed as, $\gamma = \frac{\sigma_p}{\sigma_0}$
- μ_0, σ_0, μ_p, and σ_p are the mean of observed, the mean of model outputs, the standard deviation of observed, and the standard deviation of model outputs, respectively.

Kling Gupta Efficiency (KGE)

- The KGE is interpreted relative to the mean of observations. It can range between $-\infty$ to $+1$.
- Values closer to one correspond to superior performance of the model, while lower values indicate poor performance.
- The value of $KGE > -0.41$ indicates the model is better than using the mean observations as model outputs. Thus, a value of KGE less that -0.41 may be considered as unacceptable.
- As a general guideline, $KGE = 0$ may be used as a threshold for acceptable model performance.
- Since KGE combines several performance aspects into a single measure, poor performance in one component may influence the overall metric even when other components are well represented.

Mean Absolute Error (MAE)

- The MAE is another measure of model performance that is based on the absolute error values.

$$MAE = \frac{\sum_{i=1}^{n} |O_i - P_i|}{\mathrm{n}}$$

where, $O_i (i = 1,2, \dots, n)$ is the observed values and $P_i (i = 1,2, \dots, n)$ are the corresponding model outputs.

- Unlike previous metrics, MAE has unit, which is same as the data and it can vary from 0 to ∞.
- Lower values indicate better model performance. Still, there is no threshold to indicate 'acceptable' model performance.
- Normalizing the metric to some threshold may help in better interpretation and non-dimensionalizing the metric.

Relative Absolute Error (RAE)

- Relative Absolute Error (RAE) measures the total absolute error of a model, relative to what the total absolute error would have been if the forecast had simply been the mean of the observed values.

$$RAE = \frac{\sum_{i=1}^{n}|O_i - P_i|}{\sum_{i=1}^{n}|O_i - \bar{O}|}$$

where, O_i and P_i are i^{th} value of observed and model output, respectively; and $\bar{O}$ is the mean value of the observed data.

- The RAE is also a non-negative metric with no upper bound. A value of zero indicates a perfect model.
- It represents a ratio of the level of overall agreement between the observed and model outputs.

Percent Bias $(PBIAS)$

- Percent Bias $(PBIAS)$ is a statistical performance measure used to quantify the average underestimation or overestimation of observed values by a model.
- It is expressed as a percentage, and it measures the relative magnitude of the bias between observed data and predicted data.
- The $PBIAS$ is estimated as the ratio of the average difference between the observed and the predicted data to the observed data multiplied by 100.

$$PBIAS = \frac{\sum_{i=1}^{N}(O_i - P_i)}{\sum_{i=1}^{N} O_i} \times 100$$

- Positive $PBIAS$ values represents overall overestimation by the model, whereas negative $PBIAS$ values indicate an overall underestimation by the model.

Percent Bias $(PBIAS)$

- Since $PBIAS$ is expressed relative to the observed data values, it is mainly utilized to compare the model bias across different datasets with different magnitudes.
- However, $PBIAS$ does not provide information on the variability or spread of errors and does not reflect the error distribution.
- Thus, it should be evaluated together with other performance metrics such as MAE or $RMSE$.

$$MAE = \frac{\sum_{i=1}^{N} |O_i - P_i|}{N}$$

$$RMSE = \sqrt{\frac{\sum_{i=1}^{N} (O_i - P_i)^2}{N}}$$

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Anomaly Correlation (AC)

- Sometimes we need to compare the performance of the model outputs as a map.
- The Anomaly Correlation (AC) is used as one of such performance metrics that considers the spatial variation among the adjacent grid points in the forecast and observed fields, i.e., the comparison is made between the map of observed and model output variable.
- The concept remains the same as the Pearson correlation coefficient, but it is calculated with the anomaly values, i.e., deviations from the climatological mean at each grid point or pixel.

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Anomaly Correlation (AC)

- The Anomaly Correlation (AC_C) can be expressed as

$$AC_C = \frac{\sum_{m=1}^{M}(O' - \overline{O'})(P' - \overline{P'})}{[\sum_{m=1}^{M}(O' - \overline{O'})^2 \sum_{m=1}^{M}(P' - \overline{P'})^2]^{\frac{1}{2}}}$$

where, O' and P' are the anomaly values of observed and model outputs; $\overline{O'}$ and $\overline{P'}$ are the spatial mean of the respective anomaly values; The number of grit points or pixels in the map is denoted by M.

- The magnitudes of O' and P' are calculated by subtracting the observed and model outputs from the respective climatological mean, i.e., $O' = O - \bar{C}$ and $P' = P - \bar{C}$, where $\bar{C}$ is the climatological mean.



Anomaly Correlation (AC)

- Another form of anomaly correlation is computed based on the O' and P' only, i.e., the spatial means (i.e., $\overline{O'}$ and $\overline{P'}$) of the respective anomaly are not subtracted.
- This form is known as uncentered Anomaly Correlation (AC_U), and is expressed as

$$AC_U = \frac{\sum_{m=1}^{M} O'P'}{[\sum_{m=1}^{M}(O')^2 \sum_{m=1}^{M}(P')^2]^{\frac{1}{2}}}$$

- AC_C and AC_U both yield same values only if $\sum_{m=1}^{M}(O - \bar{C}) = 0$ and $\sum_{m=1}^{M}(P - \bar{C}) = 0$.
- However, for the small size maps, this may not be possible and AC_c and AC_U may vary a little for small spatial extent of the map/field.



Performance Evaluation Metrics: Categorical Outputs

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Performance Evaluation Metrics: Categorical Outputs

Binary Classification

- For a binary classification problem, predictions lead to four possible outcomes: True Positive (TP), False Positive (FP), True Negative (TN), and False Negative (FN).
- Let's consider following table showing binary classification. Such tables are popularly known as contingency table. These tables are also possible multi-class classification.

Observation → Model output ↓	Class-1 (Positive)	Class-2 (Negative)
Class-1 (Positive)	True Positive (TP)	False Positive (FP)
Class-2 (Negative)	False Negative (FN)	True Negative (TN)

Performance Evaluation Metrics: Categorical Outputs

- From the contingency table, the following fundamental metrics are defined and used in the performance evaluation.

i. True Positive Rate (TPR), also known as *Sensitivity* or *Hit Rate* or *Recall* is expressed as

$$TPR = \frac{TP}{TP + FN}$$

ii. False Negative Rate (FNR), also known as *Miss Rate*, is expressed as

$$FNR = \frac{FN}{TP + FN}$$

iii. False Positive Rate (FPR), also known as *False Alarm Rate* (FAR), is expressed as

$$FPR = \frac{FP}{FP + TN}$$

Performance Evaluation Metrics: Categorical Outputs

- It may be noted that False Alarm Rate and False Alarm Ratio are not the same.

iv. True Negative Rate (TNR), also known as *Specificity*, is expressed as

$$TNR = \frac{TN}{FP + TN}$$

Also, it can be observed from the expressions that aforementioned four metrics are related as $TPR + FNR = 1$ and $FPR + TNR = 1$.

Binary Classification

- The True Positive Rate (TPR) and True Negative Rate (TNR) are reward-based performance metrics that evaluate the model's ability to correctly identify the occurrence and non-occurrence of events, respectively.
- Higher values indicate better model performance.
- These metrics are particularly important in applications such as rainfall forecasting, where correctly identifying rainfall and no-rainfall conditions is important for agricultural planning, and other societal and environmental decision-making.

Binary Classification

- Conversely, the False Positive Rate (FPR) and False Negative Rate (FNR) quantify forecasting errors and should ideally be minimized.
- These metrics quantify the frequency of incorrect predictions made by a model.
- They are especially useful when the objective is to reduce prediction errors, such as in temperature modelling, where forecasting errors generally have lower societal consequences than missing rainfall events.
- In addition to the above metrics, several other measures derived from the contingency table are commonly used to evaluate categorical forecast performance.
- These metrics provide additional information on different aspects of model accuracy.

Additional Performance Evaluation Metrics

- The following additional performance evaluation matrices are derived from the four possible outcomes of binary classification, i.e., True Positive (TP), False Positive (FP), True Negative (TN), and False Negative (FN).

i. Accuracy or Proportion Correct: It quantifies the accuracy of the model's predictions, since it treats correct and incorrect outcomes equally.

$$Accuracy = \frac{TP + TN}{TP + FP + TN + FN}$$

Remarks,

- Range: **0** to **1**
- Higher value indicates better performance with $Accuracy = 1$ as best performance, and $Accuracy = 0$ as worst performance

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Additional Performance Evaluation Metrics

ii. Error Rate: Error rate is expressed as the proportion of incorrect predictions out of all predictions, which is also expressed as $(1 - Accuracy)$.

$$Error\ rate = \frac{FP + FN}{TP + FP + TN + FN}$$

Remarks,

Range: **0** to **1**

- Higher value indicates poor performance with $Error\ rate = 0$ as best performance, and $Error\ rate = 1$ as worst performance

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Additional Performance Evaluation Metrics

iii. Precision: Precision quantifies the proportion of correctly predicted events among all events the model predicts. It indicates the reliability of positive predictions.

$$Precision = \frac{TP}{TP + FP}$$

Remarks,

Range: **0** to **1**

- A higher value indicates fewer false alarms.

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Additional Performance Evaluation Metrics

iv. False Alarm Ratio (FAR): FAR quantifies the proportion of the model that predicts even occurrences, but does not reflect actual occurrences. Smaller values of the FAR are preferred.

$$FAR = \frac{FP}{TP + FP}$$

Remarks,

- Range: **0** to **1**
- Lower value indicates better performance with $FAR = 0$, as the best performance and $FAR = 1$ for the worst performance.

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Additional Performance Evaluation Metrics

v. Prevalence: Prevalence quantifies the proportion of positive event occurrence (i.e., $TP + FN$), among all events (i.e., $TP + FP + TN + FN$).

$$Prevalence = \frac{TP + FN}{TP + FP + TN + FN}$$

Remarks,

- It is irrespective of the model performance in classifying the events. It has a profound, often nonlinear (quadratic) effect on predictive model performance

Additional Performance Evaluation Metrics

vi. Misclassification rate: Misclassification rate quantifies the overall frequency with which the model makes incorrect predictions, regardless of whether the error arises from falsely predicting an event or from missing an event.

$$Missclassification\ rate = \frac{FP + FN}{TP + FP + TN + FN}$$

Remarks,

- Range: **0** to **1**
- Lower values of this metric indicate better classification performance.
- A value of 0 indicates perfect accuracy, while 1 indicates total failure.

Additional Performance Evaluation Metrics

vii. Threat Score (TS) or Critical Success Index (CSI): TS or CSI quantifies the proportion of positive event detection, relative to the total number of positive event occurrences, including actually observed and model output as false positive (excluding true negative events). It helps to detect the rare events.

Remarks,

- Range: **0** to **1**
- Higher values indicates better performance with TS or $CSI = 1$ as best performance and TS or $CSI = 0$ indicates worst performance.

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Additional Performance Evaluation Metrics

viii. Odds Ratio (OR): Odds Ratio quantifies the ratio of true and false hits made by the model. It indicates the association between forecasts and observations by comparing correct and incorrect forecasts.

Remarks,

- Range: **0** to ∞
- The range $0 < OR < 1$ indicates, incorrect forecasts are more than correct forecasts.
- The $OR = 1$ indicates no association between forecasts. Where $OR > 1$ indicates correct forecasts are more than incorrect forecasts.

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Additional Performance Evaluation Metrics

ix. Bias: Bias signifies the model's performance by indicating that it predicts sufficient occurrences of events relative to the observed events.

Remarks,

- Range: **0** to ∞
- The $Bias = 1$ indicates that the model outputs and observed values with respect to positive events are equal;
- The $Bias > 1$ indicates overprediction (more positive events are predicted than observed)
- The $Bias < 1$ indicates under-prediction (fewer positive events are predicted than observed

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Additional Performance Evaluation Metrics

x. Heidke Skill Score (HSS): The HSS compares the prediction with random prediction (by chance) and evaluates the proportion of correct forecasts

$$HSS = \frac{2(TP \times TN - FP \times FN)}{(TP + FN)(FN + TN) + (TP + FP)(FP + TN)}$$

Remarks,

- Range: $-\infty$ to **1**
- The $HSS = 1$ indicates perfect prediction, while $HSS = 0$ indicates no skill compared to a random prediction.
- The $HSS < 0$ indicates performance worse than random forecast.

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Additional Performance Evaluation Metrics

xi. Peirce Skill Score (PSS): The PSS represents the difference between the TPR and FPR. The PSS is similar to HSS, except predictions are assessed for the rare events, as PSS increases for correct non-occurrences and decreases for incorrect occurrences.

$$PSS = \frac{(TP \times TN - FP \times FN)}{(TP + FN) \times (FP + TN)}$$

Remarks,

- Range: -1 to 1
- The $PSS = 1$ indicates perfect prediction, while $PSS = 0$ indicates no skill compared to a random prediction.
- The $PSS < 0$ indicates performance worse than random forecast.

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Additional Performance Evaluation Metrics

xii. Clayton Skill Score (CSS): The CSS quantifies the difference between the true hits and false hits (TP and FN).

$$PSS = \frac{(TP \times TN - FP \times FN)}{(TP + FN) \times (FN + TN)}$$

Remarks,

- Range: 0 to 1
- The $CSS = 0$ indicates model outputs do not show any improvement over random predictions.
- The $0 < CSS < 1$ indicates correct predictions of occurrence events exceed missed events.
- Higher values indicate better performance.

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Additional Performance Evaluation Metrics

xiii. Gilbert Skill Score(GSS) or Equitable Threat Score (ETS):The GSS is an improved version of TS, where random prediction is considered through TP_{ref}.

$$GSS = \frac{(TP - TP_{ref})}{TP - TP_{ref} + FP + FN}$$

where, $TP_{ref} = \frac{(TP+F\ )\times(TP+FN)}{TP+FP+FN+TN}$

Remarks,

- Range: $-\frac{1}{3}$ to 1, where $-\frac{1}{3} < GSS < 0$ indicates model performance worse than the reference prediction.
- The $GSS = 0$ indicate no improvement over reference prediction

Additional Performance Evaluation Metrics

xiv. Odds Ratio Skill Score ($ORSS$) or Yule's Q: The $ORSS$ indicates the alignment of the model occurrences, correct hits (including occurrences and non-occurrences). The $ORSS$ provides a normalized measure of association between forecasts and observations

$$ORSS = \frac{(TP \times TN) - (FP \times FN)}{(TP \times TN) + (FP \times FN)}$$

Remarks,

- Range: -1 to 1, when $ORSS = 1$, indicating Perfect agreement between observation and model output.
- The $ORSS = 0$, indicating no association between observation and model output. When $ORSS < 0$, it indicates the model predictions do not match the actual observations.


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Statistical and Machine Learning Methods: Basics and Applications

MODULE 8 : Statistical Learning

LECTURE W8L4: Concept of Regularization

BY

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CONCEPTS COVERED

Regularization in Regression

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Regularization

- Regularization is a variety of techniques used to enable a model to neither underfit nor overfit the data and thus the model learns (train) properly from the given data.
- The model error over test data (i.e., data not used in training the model) is called the *generalization error,* and error over training data is called as *training error.*
- Regularization techniques aim to minimize the *generalization error* of the model without having an impact on the training error.

Why Regularization?

- A major problem with statistical or machine learning methods is that often a large number of predictors are available, but only a fraction of them are actually significant, resulting in overfitting of the model.
- The way of solving this issue that divides the statistics and machine learning methods.
- Statistical models tend to delete some predictors, making the model simpler and more interpretable.
- However, when two predictors are highly correlated, a selection method may choose one and discard the other.
- Small changes in the sample can reverse this choice, making predictor selection unstable and not always reproducible.

Why Regularization?

- Also, discarding the predictors is equivalent to discarding the information, as predictor set variables give unique information.
- The set of techniques called regularization techniques is used to prevent overfitting, although the predictor set is large.
- Regularization refers to a variety of techniques used in models to solve ill-posed problems or to prevent overfitting.
- In the statistics context of multivariate linear regression, it will lead to overfitting for a large predictor set.
- To alleviate the overfitting, among the regularization techniques, the shrinkage methods are reliable.
- The shrinkage methods only shrink the regression parameters towards zero.

Ridge Regression

- The regularization applied to multivariate linear regression is called ridge regression.
- To estimate the regression parameters in multivariate linear regression, the squared sum of errors (SSE). (to be indicated in particular to MLR)

$$SSE = \sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{m} \beta_j x_{ij} \right)^2$$

- In ridge regression, the objective function (J) is to be minimized, i.e., SSE plus a regularization term.

Ridge Regression

- In scalar form, the objective function (J) is as follows:

$$J = \sum_{i=1}^{N}\left(y_i - \beta_0 - \sum_{j=1}^{m}\beta_j x_{ij}\right)^2 + \lambda\sum_{j=1}^{m}\beta_j^2 = SSE + \lambda\sum_{j=1}^{m}\beta_j^2$$

where $\lambda \geq 0$ is a regularizing parameter to reduce the overall (J).

- The regularization term suppresses the magnitude of the weight or regression parameters β_j, hence , the term is also called a weight penalty.
- There is no suppression on the magnitude of β_0 since the y-intercept cannot be forced to shrink towards 0.
- The standardization is considered an essential step for the weight penalty term to work evenly on the predictors.

Ridge Regression

- The mean of standardized predictors will give $\beta_0 = \bar{y}$, where $\bar{y}$ is the mean of y.
- Therefore, for the centered y, β_0 is no longer present in the objective function.
- The objective function J in vector form can be expressed as:

$$J = (\boldsymbol{y} - \boldsymbol{X\beta})^{\mathbf{T}}(\mathbf{y} - \mathbf{X\beta}) + \lambda\boldsymbol{\beta^T\beta}$$

- From the MLR solution, the ridge regression solution is,

$$\hat{\beta} = \left(\boldsymbol{X^TX} + \lambda\boldsymbol{I}\right)^{-1}\boldsymbol{X}^T\boldsymbol{y}$$

- The new term $\lambda\boldsymbol{I}$ adds a positive constant to the diagonal of $\boldsymbol{X}^T\boldsymbol{X}$.

Advantages of Ridge Regression

- Many a times the matrix $\boldsymbol{X^T X}$ will be nearly singular due to the strong correlation between some predictors.
- Small errors in the elements of $\boldsymbol{X}$ lead to large changes in the computed $(\boldsymbol{X^T X})^{-1}$, therefore, these conditions of $\boldsymbol{X^T X}$ are called ill-conditioned.
- Adding $\lambda \boldsymbol{I}$ improves the numerical stability when computing the inverse of $\boldsymbol{X^T X}$.
- The optimal value for λ is estimated by cross-validation.
- Models are trained using the trained data for a number of specified λ values, and the model performance over the validation data is then used to select the optimal λ value.

Determination of λ

- Alternatively, to minimize the objective function (J) is to have the weight penalty expressed as an explicit constraint, that is

$$J = \sum_{i=1}^{N}\left(y_i - \beta_0 - \sum_{j=1}^{m}\beta_j x_{ij}\right)^2 \text{, subject to } \sum_{j=1}^{m}\beta_j^2 \leq b.$$

- From the Lagrange multipliers, an explicitly constrained optimization problem as shown in above equation is equivalent to an unconstrained optimization problem containing a Lagrange multiplier term i.e.,

$$J = \sum_{i=1}^{N}\left(y_i - \beta_0 - \sum_{j=1}^{m}\beta_j x_{ij}\right)^2 + \lambda\sum_{j=1}^{m}\beta_j^2$$

- The exact relation between the constraint bound b and the Lagrange multiplier λ is data independent.

Example

For the given training and validation data, perform ridge regression and find the optimal lambda for the reduced MSE.

X_{train}	Y_{train}	$X_{validation}$	$Y_{validation}$
6.90	5.40	4.00	4.0
6.40	4.10	5.00	4.50
6.50	4.70	6.00	5.00
5.10	3.50	7.00	5.50
7.10	4.80	8.00	6.00
7.10	5.00	9.00	6.50
10.20	8.20	10.00	7.00
9.90	7.00	11.00	7.50
8.40	5.70		
5.80	7.40		

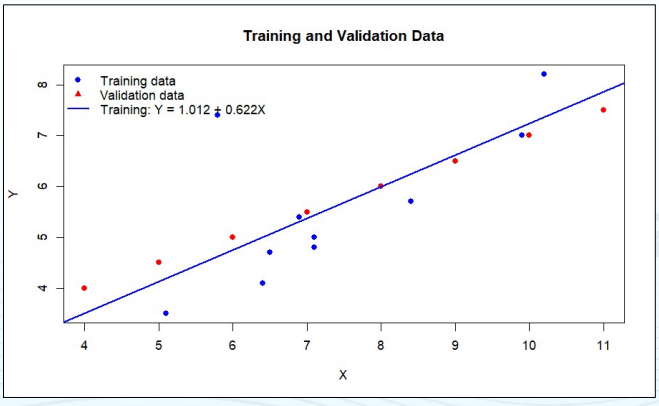
Given, from least square regression for training data, the S_{xx} and S_{xy} are calculated as

$$S_{xx} = 15.65$$
$$S_{xy} = 25.14$$

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Solution

Visualization of the data in a scatter plot



Fitted Equations through least square regression

- $y_{train} = 1.012 + 0.622x_{train}$

Solution

For ridge regression, the constrained problem is:

$$J = \sum_{i=1}^{N}\left(y_i - \beta_0 - \sum_{j=1}^{m}\beta_j x_{ij}\right)^2 \text{, subject to } \sum_{j=1}^{m}\beta_j^2 \leq b.$$

Introducing a Lagrange (KKT) multiplier $\lambda \geq 0$ for the inequality constraint gives the Lagrangian

$$L(\beta_0, \beta, \lambda) = \sum_{i=1}^{N}\left(y_i - \beta_0 - \sum_{j=1}^{m}\beta_j x_{ij}\right)^2 + \lambda\left(\sum_{j=1}^{m}\beta_j^2 - b\right)$$

$$\frac{\partial L}{\partial \beta_0} = 0$$

$$\frac{dL}{d\beta_j} = -\sum x_i y_i + \beta_j \sum x_i^2 + \lambda\beta_j = 0$$

Solution

Therefore, on differentiating with respect to β_j and β_0 as follows:

$$\frac{\partial L}{\partial \beta_0} = 0$$

$$\frac{dL}{d\beta_j} = -\sum x_i y_i + \beta_j \sum x_i^2 + \lambda\beta_j = 0$$

$$\Rightarrow \sum x_i y_i = \beta_j\left(\sum x_i^2 + \lambda\right)$$

$$\beta_j = \frac{\sum x_i y_i}{\left(\sum x_i^2 + \lambda\right)} \Rightarrow \frac{S_{xy}}{S_{xx} + \lambda}$$

- From the given data, $j = 1$

$$\beta_1 = \frac{15.65}{25.14+\lambda} \text{ and } \beta_0 = \bar{y} - \beta_1(\lambda)\bar{x}$$

Solution

Therefore, the intercept is,

$$\beta_0(\lambda) = \frac{1}{8}\left(\sum y - \beta_1(\lambda)\sum x\right) \Rightarrow 5.58 - 7.34\beta_1(\lambda)$$

Validation prediction (for a validation point x_v),

$$y_v(\lambda) = \bar{y} + \beta_1(\lambda)(x_v - \bar{x})$$

Let, $z_v = x_v - \bar{x}$, $d_v = y_v - \bar{y}$

Then, the validation MSE is as follows,

$$MSE_{val}(\lambda) = \frac{1}{8}\left(\sum_{v=1}^{8} d_v - a_1(\lambda)z_v\right)^2$$

Solution

- On using the validation data, with $\bar{x} = 7.34$ and $\bar{y} = 5.58$,

$$\sum z_v^2 = 42, \sum z_v d_v = 21, \sum d_v^2 = 10.5$$

$$MSE_{val}(\lambda) = \frac{1}{8}\left(\sum_{v=1}^{8} d_v - a_1(\lambda)z_v\right)^2 \Rightarrow \sum d_v^2 - 2a_1\sum z_v d_v + a_1^2\sum z_v^2$$

$$\Rightarrow MSE_{val}(\beta_1) = \frac{1}{8}(42\beta_1^2 - 2(21)\beta_1 + 10)$$

On differentiating MSE_{val} with respect to β_1 and equating to zero we get,

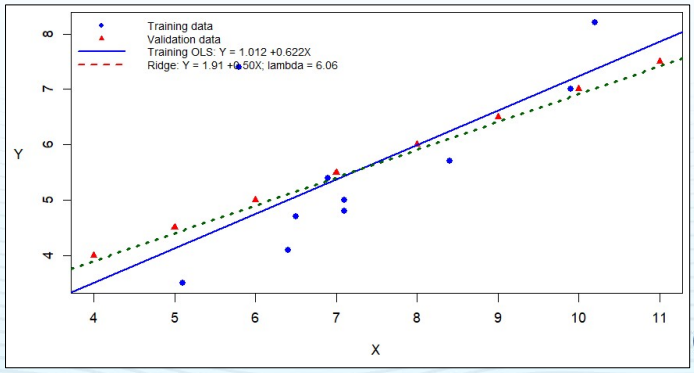
$$\beta_1 = 0.50,$$

- Therefore,

$$\beta_1 = \frac{15.65}{25.14 + \lambda} \Rightarrow 0.50 = \frac{15.65}{25.14 + \lambda} \Rightarrow \lambda \approx 6.06$$

Solution

- The optimal ridge model is $\beta_1 = 0.50$, and $\beta_0 = 5.58 - 7.34(0.50) \approx 1.91$
- Therefore, the ridge regression equation as follows, $\bar{y} = 1.91 + 0.5x$
- The slope parameter reduced from 0.622 from training equation to 0.50 in ridge regression.



Importance of Ridge Regression

- The explicit constraint ensures that the regression parameters are confined within a hypersphere of radius b, thereby preventing the overfitting from using the large coefficients ($|\beta_j|$).
- The ridge regression is suitable for datasets with a small length and a larger number of predictors.
- However, the ridge regression outperformance over MLR disappears when the sample size greatly exceeds the number of regression parameters.
- Contrastingly, the weight penalty term introduces a bias in the solution.

Lasso regression

- LASSO (Least Absolute Shrinkage and Selection Operator) is a linear regression method that simultaneously penalizes coefficient values and selects the most relevant predictors.
- Along with increased prediction accuracy, the model interpretability is also enhanced.
- The standardization steps are similar to ridge regression, and while imposing the constraint for the minimization of the objective function J as follows,

$$J = \sum_{i=1}^{N}\left(y_i - \beta_0 - \sum_{j=1}^{m}\beta_j x_{ij}\right)^2 \text{, subject to } \sum_{j=1}^{m}|\beta_j| \leq b$$

Lasso regression

- The alternative Lagrange form converts the constraint to a weight penalty term in the objective function, that is,

$$J = \sum_{i=1}^{N}\left(y_i - \beta_0 - \sum_{j=1}^{m}\beta_j x_{ij}\right)^2 + \lambda\sum_{j=1}^{m}|\beta_j|$$

where, λ is a Lagrange multiplier.

- The difference between lasso and ridge regression is that while ridge regression can make some regression parameters or weights small, it does not set them to zero.
- In lasso, the coefficients are set to zero, which is equivalent to removing the predictor effect on the response.


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Statistical and Machine Learning Methods: Basics and Applications

MODULE 8 : Statistical Learning

LECTURE W8L5: Concept of Regularization

BY

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CONCEPTS COVERED

Regularization in Machine Learning

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Regularization

- Regularization techniques are very important in neural networks and other machine learning models to prevent overfitting.
- A similar technique of adding a weight penalty (also called weight decay in machine learning approaches) to the objective function is followed.
- If MSE is the objective function for any of the machine learning algorithms, the weight penalty is added to it as follows,

$$J = \frac{1}{N}\sum_{n=1}^{N}\left\{\frac{1}{2}\sum_{p}\left[\hat{y}_p^n - y_p^n\right]^2\right\} + \frac{\lambda}{2}\sum_{j=1}^{N_w} w_j^2$$

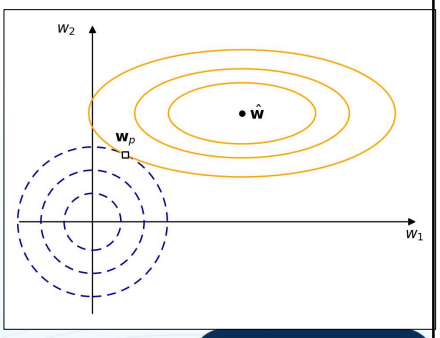
The second term is the weight penalty term, $\lambda \geq 0$ is the weight penalty parameter. This is also known as weight decay parameter or regularization parameter.

Weight Penalty

- The weight penalty term represent a parabolic surface of N_w dimensions, where N_w is the number of weights present in the algorithm.
- The parameter λ is considered as the hyperparameter, as it exerts control over the weight and offset parameters of the machine learning algorithm.
- During the non-linear optimization, the λ is held constant and searched for other optimal values.
- The positive values of λ and selection of larger $|w_j|$ during the optimization would increase the value of J, due to these larger values of $|w_j|$ values are penalized
- The selection of larger λ will increase the suppression on the magnitude of the weights during the search for the optimal weights.

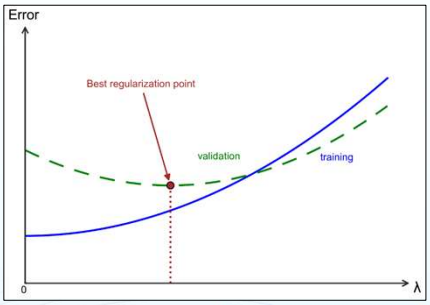
Weight penalty

- The w_1 and w_2 are the weights that are supposed to be optimized in a two-dimensional parabola.
- The vector of weights **$(\widehat{w})$** found without using the weight penalty and the optimal **(w_p)** found with the weight penalty in the objective function (J).
- The magnitudes of both w_{p1} and w_{p2} from the $\boldsymbol{w_p}$ have smaller magnitudes than $\widehat{w_1}$ and $\widehat{w_2}$, respectively.
- The weight penalty reduces the magnitude of less sensitive weights more, than the sensitive weights.
- Therefore, the weight penalty reduces the magnitude of $\widehat{w_1}$ much strongly than $\widehat{w_2}$, i.e., $|w_{p1}| \ll |\widehat{w_1}|$ and $|w_{p2}| < |\widehat{w_2}|$.



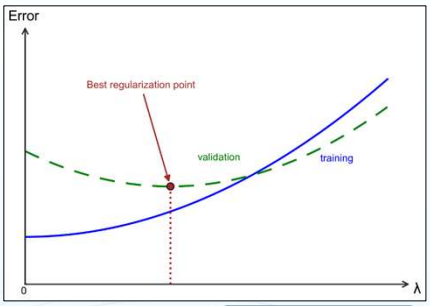
Determination of weight penalty

- The common method to select the λ is by validation, by dividing the available data into two subsets, namely training data and validation data.
- Models trained for divided training data for the specified set of λ values.
- The model error on the training data increases as the λ increases; however, the model error over the validation data gradually reduces, and eventually rises for a small increase in the λ.
- The increase in the validation error is inferred as an excessively flexible model has started to overfit the training data.



Determination of weight penalty

- To ascertain the optimal weight penalty value (λ), the λ values are used in logarithmic scale (i.e., 10^{-5}, 10^{-4}, 10^{-3}, 10^{-2} and 10^{-1} etc.).
- The run with smallest validation error is the optimal value of λ.
- Apart from these, many other approaches are prevalent and popular such as maximum norm constraint approach, since the advent of deep neural networks.



Early Stopping

- Early stopping is also a simple regularization of the technique, which is commonly used either alone or in conjunction with weight penalty.
- In a similar way as in case of the weight penalty, the available data is divided into training and validation data.
- The training and validation are carried out at each epoch, and the objective function is evaluated over the validation data, which decreases to a minimum, then increases due to overfitting.
- The minimum validation error gives an indication to stop the training at that epoch, as additional training epochs only contribute to overfitting.
- The optimal number of training epochs (N_e) is determined, and training and validation are carried out for the N_e epochs.

Early Stopping

- Here, N_e is a hyperparameter.
- In general, multiple runs are required to ascertain the optimal value of any hyperparameter by validation.
- However, a single run is needed to ascertain the optimal value of N_e.
- An extra memory needed of course.
- With this extra memory, the model parameters and weights are save for each run.
- After the validation error is noticed to exhibit an increasing trend, we need to go back and recover the model at the best performing epoch, which is denoted as the optimum epoch (N_e^{th} epoch).

Cross-Validation

- If the dataset is large, it is convenient to reserve some data for validation purposes.
- In contrast, in the case of a small dataset, it is not possible to reserve a fair amount of data for validation, since there are fewer observations for model training.
- On the other hand, even with the availability of a large dataset, the validation error on a single set of data is unreliable.
- Cross-validation is a technique that allows the entire dataset to be used for validation.

Cross-Validation

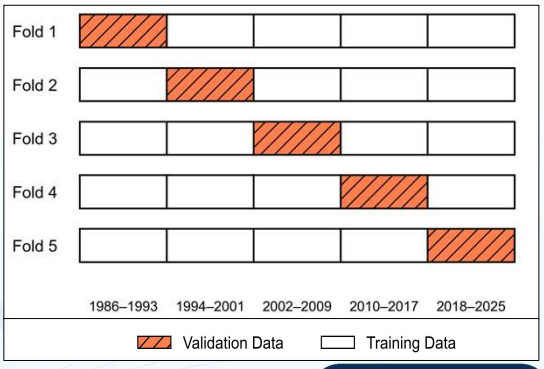
- From the available data, the cross-validation process starts with dividing the data into approximately equal segments.
- If the data is divided into K equal parts, one segment is reserved as validation data, while the other $K-1$ segments are training data used for model training.
- This process is repeated for K times, so that each segment of the data record has been used as validation data.
- The validation performance score (e.g. the MSE or other) is computed using every data point in the data record.

Cross-Validation

- The model is made to run for different values of tuning parameters (e.g., weight penalty, number of training epochs, etc.)
- Based on the best validation performance score over the whole data record, one can obtain the optimal values of tuning parameters (also called hyperparameters).
- The resulting optimal tuning parameters are usually used to train the model again using all available data to obtain the final model.
- This process is repeated for K times, so that each segment of the data record has been used as validation data.

Cross-Validation

Consider a continuous 40-year historical dataset spanning 1986 to 2025, used to train and validate a predictive hydro-meteorological model for streamflow prediction. To rigorously evaluate the performance while accounting for temporal dependencies, a 5-fold cross-validation scheme is implemented by dividing the timeline into five blocks of 8 years each.



To optimize model complexity, a grid search is performed over five regularization penalty (λ) values. The model is trained and validated across all folds for each λ, and performance is evaluated by averaging the validation Root Mean Squared Error (RMSE) and for optimal epochs required.

Cross-Validation

The aggregated results are as follows:

Penalty (λ)	Average Optimal Epochs	Average validation RMSE (m^3/s)
0.0001	120	30.45
0.001	65	25.85
0.01	40	30.10
0.1	25	42.86
1.0	10	55.36

Suggest the best possible weight penalty (λ), and the number of training epochs is to be considered for the reduced value of $RMSE$ of the model.

Cross-Validation

Optimal hyperparameter selection:

- Optimal Penalty (λ): 0.001
- Average optimal epochs: 65
- Average validation $RMSE$: 25.85 m^3/s
- As the penalty (λ) increases from 0.0001 to 1, the average optimal training epochs decrease steadily from 120 to 10.

- The validation $RMSE$ initially drops to minimum of 25.85 m^3/s at $\lambda = 0.001$
- Then $RMSE$ value increased up to 55.36 m^3/s illustrating the transitioning from controlling the model variance (overfitting) at lower penalties to introducing excessive bias (underfitting) at higher penalties.

Points to remember on cross-validation

- The time series data are often not independent of each other due to serial correlation.
- For example, the mean streamflow in April is be dependent on the March streamflow.
- Due to the serial correlation, making part of the data independent validation data is not encouraged, since it is correlated with neighboring observations already used for training.
- The validation segments need to be longer than the timescale in which there is serial correlation.

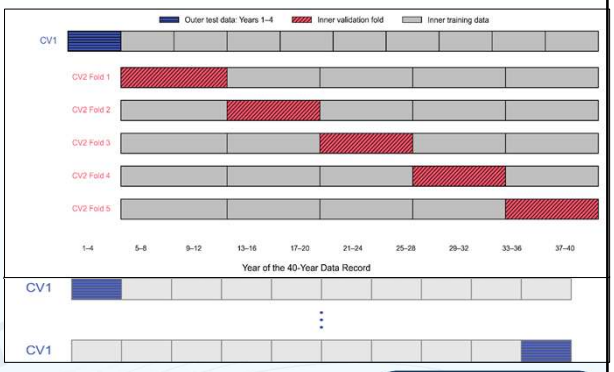
Points to remember on cross-validation

- If there is a correlation between the data immediately at the initial part of the validation data, then the correlated part of the validation data is to be included in the training data.
- Otherwise, the serial correlation can lead to an underestimation of the model error over the validation data, especially when using small validation segments.
- Although in the presence of serial correlation, it is assumed that the best way to perform K-fold cross validation through leave-one-out validation.
- But, due to the presence of serial correlation, it is the method most susceptible to spurious skills caused by serial correlation.

Illustrating the effect of serial correlation

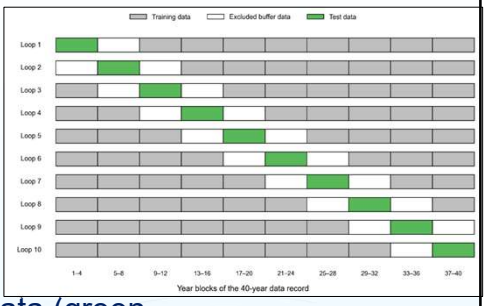
- The validation data is used for model selection (tuning parameters, weights, and penalties). Therefore, the validation error does not represent the true out-of-sample forecast error.
- True prediction error must be evaluated using completely independent data that was never touched during the model training or validation.
- Therefore, the data record must be divided into three separate sets: training, validation, and testing data.
- The method called double cross-validation (also known as nested cross-validation) is used to achieve the rigorous separation without wasting data.
- Even though it is computationally expensive, it is highly effective for unbiased estimation.

Double cross validation



- Steps: Let us assume that for 40 years of continuous data, 1-4 years of data is reserved as test data.
- The rest of the data (5 to 40) is used for training and validation.
- Then implementation of the K-fold cross-validation solely on years 5 to 40 to find the hyperparameters, then test those predictions on years 1 to 4.
- In a similar way, shift the test window, e.g., reserve years 5 to 8 for testing and use 1 to 4 and 9 to 40 for training and validation.
- Repeat this rotation until an error is computed across the entire data record.

Addressing the Boundary issues



- To tackle the serial correlation challenge across boundaries, standard double cross-validation can be adapted.
- Implementing a modified double cross-validation scheme helps further eliminate artificial skill introduced by adjacent data overlaps.
- As shown, the training data (grey) and the test data (green portion).
- The data segments (in white) bridging the training data and test data are not used to avoid serial correlation.
- The test data segment is moved repeatedly for a complete data record over to the end in this cross-validation loop, so forecast performance is tested over the whole record.


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